

# README

## 1 AI Standards & Protocols Knowledge Base

**Version:** 1.0  
**Generated:** December 4, 2025  
**Purpose:** Comprehensive reference documentation for AI/ML protocols, formats, and integration standards

### 1.1 Overview

This knowledge base provides in-depth documentation of AI-related protocols, standards, and formats used across the machine learning and artificial intelligence ecosystem. It covers neural network fundamentals, model interchange formats, API integration standards, serving protocols, ML lifecycle management, and related technologies.

### 1.2 Structure

```
ai-standards-kb/
├── INDEX.md                # Main navigation and overview
├── README.md              # This file
├── generate-pdfs.ps1      # PDF generation script
├── PDF_CONVERSION_INSTRUCTIONS.txt # Manual conversion guide
├──
├── standards/             # Core AI standards documentation
│   ├── 01-Neural-Networks-Fundamentals.md
│   ├── 02-ONNX.md
│   ├── 03-OpenAPI-AI-Services.md
│   ├── 04-Model-Cards.md
│   ├── 05-TensorFlow-Serving.md    # [To be completed]
│   ├── 06-MLflow.md                # [To be completed]
│   ├── 07-Hugging-Face-Hub.md      # [To be completed]
│   ├── 08-gRPC-ML-Serving.md        # [To be completed]
│   ├── 09-KServe.md                # [To be completed]
│   ├── 10-Triton-Inference-Server.md # [To be completed]
│   └── 11-Other-AI-Standards-Summary.md
├── annexes/              # Related technologies and ecosystems
│   ├── A01-Vector-Embeddings-Standards.md # [To be completed]
│   ├── A02-Vector-Databases.md            # [To be completed]
│   ├── A03-RAG-Patterns.md                # [To be completed]
│   ├── A04-Prompt-Engineering-Standards.md # [To be completed]
│   └── A05-LLM-API-Standards.md            # [To be completed]
├──
├── pdf/                   # Generated PDF documents
│   ├── INDEX.pdf
│   ├── standards/
│   └── annexes/
│       # PDF versions of standards
│       # PDF versions of annexes
```

### 1.3 Current Status

#### 1.3.1 Completed Documents (4/16)

- ☒ **INDEX.md** - Complete navigation and overview
- ☒ **01-Neural-Networks-Fundamentals.md** - Abstract NN design, training, inference (12,000+ words)
- ☒ **02-ONNX.md** - ONNX format specification and usage (7,500+ words)
- ☒ **03-OpenAPI-AI-Services.md** - OpenAPI for AI REST APIs (7,000+ words)
- ☒ **04-Model-Cards.md** - Model documentation framework (6,500+ words)
- ☐ **11-Other-AI-Standards-Summary.md** - Summary of 14 additional standards (6,000+ words)

#### 1.3.2 In Progress / To Be Completed (11/16)

- ☐ **05-TensorFlow-Serving.md** - TF Serving protocols and APIs

- ☐ **06-MLflow.md** - ML lifecycle management platform
- ☐ **07-Hugging-Face-Hub.md** - Model sharing and deployment
- ☐ **08-gRPC-ML-Serving.md** - gRPC patterns for ML
- ☐ **09-KServe.md** - Kubernetes ML serving
- ☐ **10-Triton-Inference-Server.md** - NVIDIA inference platform
- ☐ **A01-Vector-Embeddings-Standards.md** - Embedding formats and metrics
- ☐ **A02-Vector-Databases.md** - Vector DB specifications
- ☐ **A03-RAG-Patterns.md** - RAG architectures
- ☐ **A04-Prompt-Engineering-Standards.md** - Prompt templates
- ☐ **A05-LLM-API-Standards.md** - LLM API patterns

## 1.4 ☐ Quick Start

### 1.4.1 Reading the Documentation

1. **Start with the INDEX:** Open `INDEX.md` for a complete overview and navigation
2. **Follow the links:** Documents are cross-linked for easy navigation
3. **Use the PDF versions:** Generate PDFs for offline reading (see below)

### 1.4.2 Generating PDFs

#### 1.4.2.1 Option 1: Using the PowerShell Script

```
cd ai-standards-kb
.\generate-pdfs.ps1
```

The script will: - Detect all markdown files - Check for Pandoc installation - Offer automated conversion if Pandoc is available - Generate instructions for manual conversion

#### 1.4.2.2 Option 2: Manual VS Code Conversion

1. Install “Markdown PDF” extension (yzane.markdown-pdf) in VS Code
2. Open each markdown file
3. Press `Ctrl+Shift+P` (Windows/Linux) or `Cmd+Shift+P` (Mac)
4. Select “Markdown PDF: Export (pdf)”
5. Move generated PDFs to appropriate `pdf/` subdirectories

#### 1.4.2.3 Option 3: Using Pandoc (Command Line)

```
# Single file
pandoc INPUT.md -o OUTPUT.pdf --pdf-engine=xelatex --toc

# Batch conversion (all standards)
Get-ChildItem standards/*.md | ForEach-Object {
    pandoc $_.FullName -o "pdf/standards/($_.BaseName).pdf" --pdf-engine=xelatex --toc
}
```

## 1.5 ☐ Document Features

Each documentation file includes:

### 1.5.1 1. Comprehensive Structure

- **Overview:** Purpose and key features
- **Authoritative References:** Official specifications and standards bodies
- **Format Structure:** Detailed entity/attribute descriptions
- **Procedural Use-Cases:** Step-by-step workflows with diagrams
- **Examples:** Both pseudo-data and real-world implementations
- **Tools & Ecosystem:** Software implementing the standard

- **Best Practices:** Guidelines and recommendations

## 1.5.2 2. Code Examples

- **Pseudocode:** For algorithmic understanding
- **C++:** For performance-critical implementations
- **Python:** For practical API usage
- **YAML/JSON:** For configuration formats

## 1.5.3 3. Visual Diagrams

- **Mermaid flowcharts:** Process flows
- **Sequence diagrams:** API interactions
- **Architecture diagrams:** System components

## 1.5.4 4. Cross-References

- Links between related documents
- References to external authoritative sources
- Navigation aids throughout

# 1.6 □ Use Cases

## 1.6.1 For ML Engineers

- Understand model interchange formats (ONNX, CoreML)
- Learn deployment patterns (TensorFlow Serving, Triton)
- Implement production APIs (OpenAPI, gRPC)

## 1.6.2 For Data Scientists

- Document models properly (Model Cards)
- Version datasets and experiments (DVC, MLflow)
- Understand embedding standards

## 1.6.3 For AI Architects

- Design AI integration patterns
- Evaluate serving platforms
- Plan ML infrastructure

## 1.6.4 For Compliance & Governance

- Understand model documentation requirements
- Implement responsible AI practices
- Meet regulatory standards

# 1.7 □ Requirements

## 1.7.1 For Reading

- Markdown viewer (VS Code, GitHub, any text editor)
- Web browser for interactive diagrams

## 1.7.2 For PDF Generation

- **VS Code** with “Markdown PDF” extension (yzane.markdown-pdf), OR
- **Pandoc** 2.0+ with xelatex engine

## 1.7.3 For Code Examples

- **Python 3.8+** for Python examples
- **C++17** compiler for C++ examples
- Relevant ML frameworks as needed

## 1.8 □ Contributing

This documentation represents the state of AI standards as of December 4, 2025. To expand or update:

1. Follow the existing document structure
2. Include all required sections (see template above)
3. Provide both abstract and concrete examples
4. Add cross-references to related documents
5. Update INDEX.md with new entries

## 1.9 □ License

This documentation aggregates information from numerous open standards and specifications. Each document contains proper attribution and links to original sources. Please refer to individual standard licenses for usage terms.

### 1.10 □ Document Information

- **Generated:** December 4, 2025
- **Version:** 1.0
- **Scope:** AI/ML protocols, formats, and integration standards
- **Coverage:** ~40,000 words across completed documents
- **Target Completion:** 60,000+ words across all documents

### 1.11 □ External Resources

#### 1.11.1 Standards Bodies

- **ONNX:** <https://onnx.ai/>
- **OpenAPI Initiative:** <https://www.openapis.org/>
- **Khronos Group (NNEF):** <https://www.khronos.org/nnef>
- **Linux Foundation AI & Data:** <https://lfaidata.foundation/>

#### 1.11.2 Tools & Platforms

- **ONNX Runtime:** <https://onnxruntime.ai/>
- **TensorFlow:** <https://www.tensorflow.org/>
- **PyTorch:** <https://pytorch.org/>
- **Hugging Face:** <https://huggingface.co/>
- **MLflow:** <https://mlflow.org/>

#### 1.11.3 Learning Resources

- **Deep Learning Book:** <https://www.deeplearningbook.org/>
- **Stanford CS231n:** <http://cs231n.stanford.edu/>
- **Fast.ai:** <https://www.fast.ai/>

---

**Next Steps:** 1. Review completed documentation in INDEX.md 2. Run `generate-pdfs.ps1` to create PDF versions 3. Check STATUS.md for implementation roadmap for remaining documents

For questions or clarifications, refer to the authoritative references linked in each document.